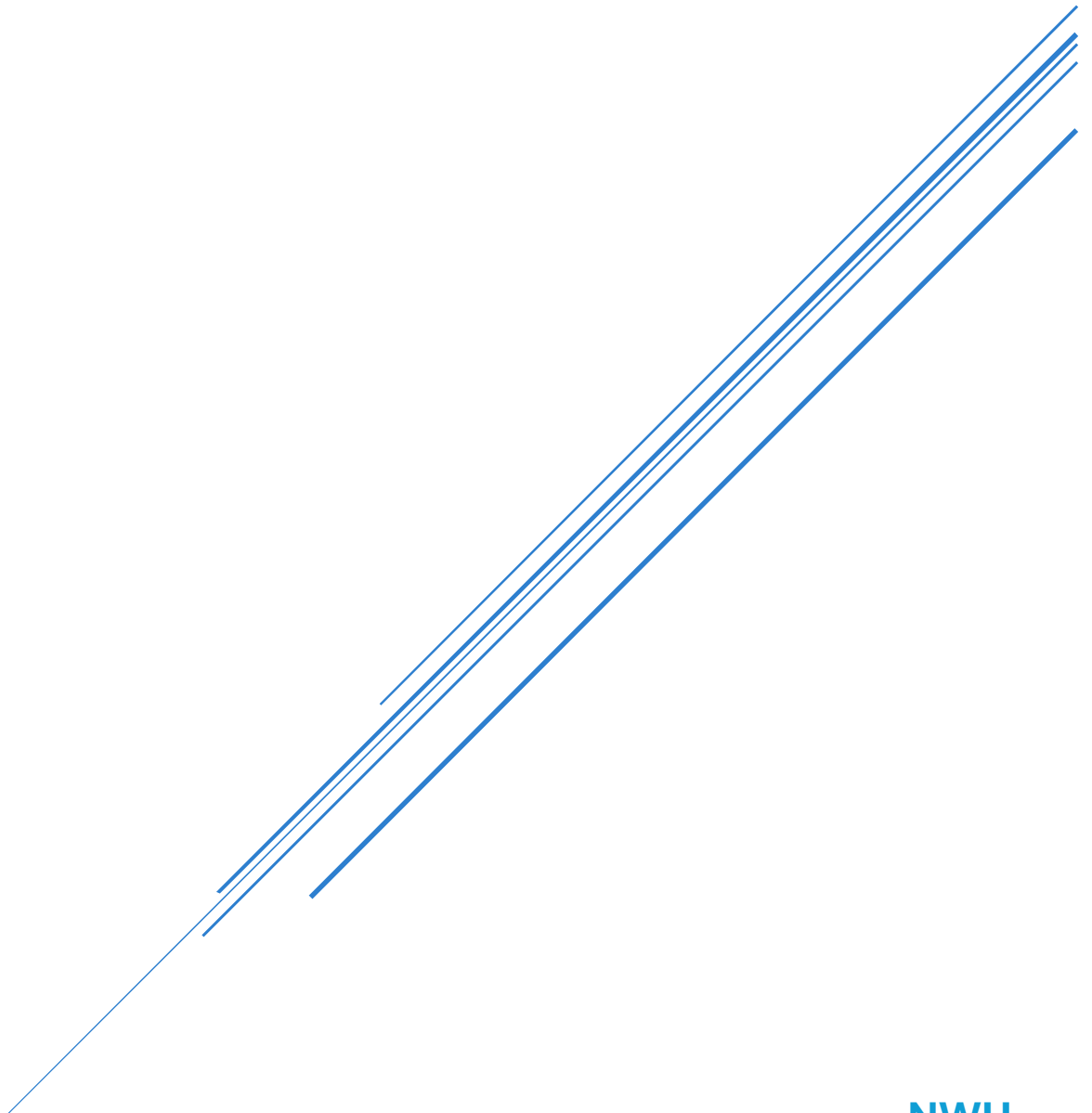


COMPUTER NETWORKS

Milestone 1:

Name and Surname: Sam Mbongela

Student Number: 43520286



1. Client Requirements

1.1 Context

1.2 Functional Requirements

1.3 Non-Functional Requirements

1.4 Requirement-to-Design Mapping

2. Physical Topology

2.1 Architecture

2.2 Devices

2.3 Connections

2.4 UPS Resilience

3. Logical Topology

3.1 VLAN Structure

3.2 Routing & Traffic Control

3.3 HTTP Server Placement

3.4 Dual-WAN Failover

3.5 Collision & Broadcast Management

4. IP Addressing Plan

4.1 VLSM Allocation

4.2 Device Assignments

4.3 WAN & Transit Networks

4.4.1 WAN & Transit Networks

4.4.2 DHCP Configuration

4.5 Firewall ACL Rules

4.6 Testing & Verification Plan

4.7 Design Summary

5. Initial GitHub repository

6. Conclusion

Project Overview:

Project ID	CMPG325-20209-067
Client ID	CLI-067
Organisation	Tiro Mining Engineering Services (Vryburg)
Industry	Engineering
Technical Challenge	Internal HTTP/Web Server Hosting
Change Request	CR15 – Dual Internet Connectivity
Design Constraint:	Load-shedding resilience, minimal device count
Implementation Platform:	Cisco Packet Tracer

1. Client Requirements:

1.1 Context

Tiro Mining Engineering Services requires a secure, resilient, and efficient network to support engineering workflows such as CAD file sharing, project documentation, and internal collaboration tools. The design must use the allocated 192.168.36.0/24 block, integrate dual ISP connections for resilience, and ensure critical devices remain operational during load-shedding.

1.2 Functional Requirements

- Provide connectivity for client workstations (PC1–PC3).
- Host an internal HTTP/Web server (SRV1) with static IP.
- Deliver Internet access via ISP1 (primary) and ISP2 (backup).
- Enable WAN failover through R1 Edge Router.
- Enforce traffic control via FW1 Firewall.
- Segment traffic into VLANs (10 Management, 20 Servers, 30 Clients).
- Support end-to-end connectivity and HTTP service testing in Packet Tracer.

1.3 Non-Functional Requirements

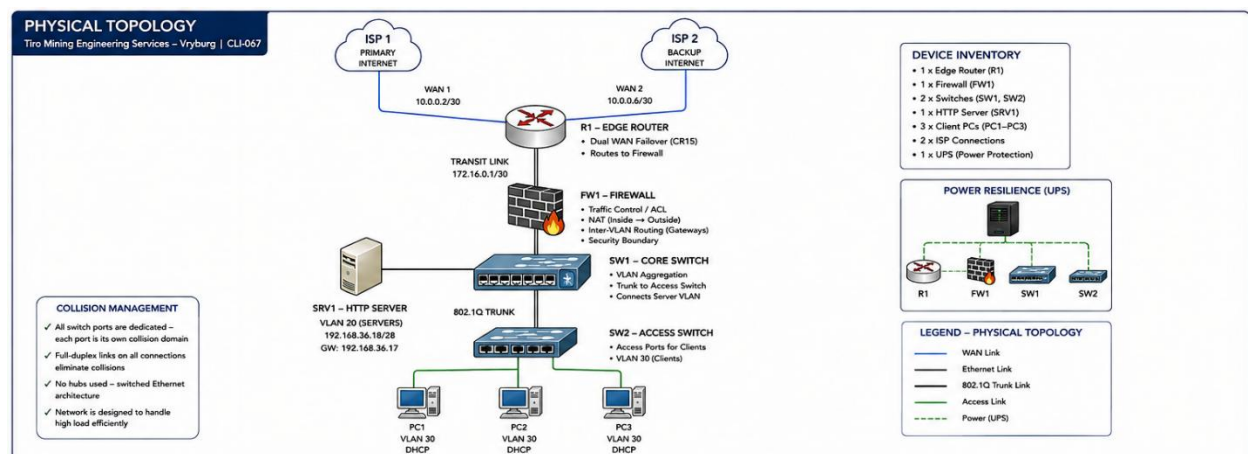
- Reliability: ISP2 backup ensures continuity if ISP1 fails.
- Resilience: UPS-backed R1, FW1, SW1 maintain uptime during load-shedding.
- Security: FW1 restricts unauthorised inbound traffic.
- Scalability: VLSM leaves unused address space for future growth.
- Manageability: VLANs and IPs follow a clear structure.
- Efficiency: Minimal device count avoids unnecessary complexity.

1.4 Requirement-to-Design Mapping:

Requirement	Design Response
Internal HTTP server	SRV1 in VLAN 20 with static IP (192.168.36.18)
Dual ISP resilience (CR15)	R1 with ISP1 primary, ISP2 backup
Traffic control	FW1 firewall with ACL/NAT
Load-shedding resilience	UPS-backed R1, FW1, SW1
Minimal device count	1 router, 1 firewall, 2 switches
Segmentation	VLANs 10, 20, 30
Client addressing	DHCP in VLAN 30
Infrastructure addressing	Static IPs for predictability

2. Physical Topology

2.1 Architecture



Compact hierarchy: **ISP1/ISP2 → R1 Edge Router → FW1 Firewall → SW1 Core Switch → SW2 Access Switch → PCs + Server**

2.2 Devices

- **ISP Connections (2):** Primary and backup Internet.
- **R1 Edge Router:** WAN connectivity and failover.
- **FW1 Firewall:** Security boundary, ACL/NAT.
- **SW1 Core Switch:** VLAN aggregation, server connection.
- **SW2 Access Switch:** Client connectivity.
- **SRV1 HTTP Server:** Internal web service.
- **PC1–PC3:** Client workstations.
- **UPS:** Power resilience for critical devices.

2.3 Connections

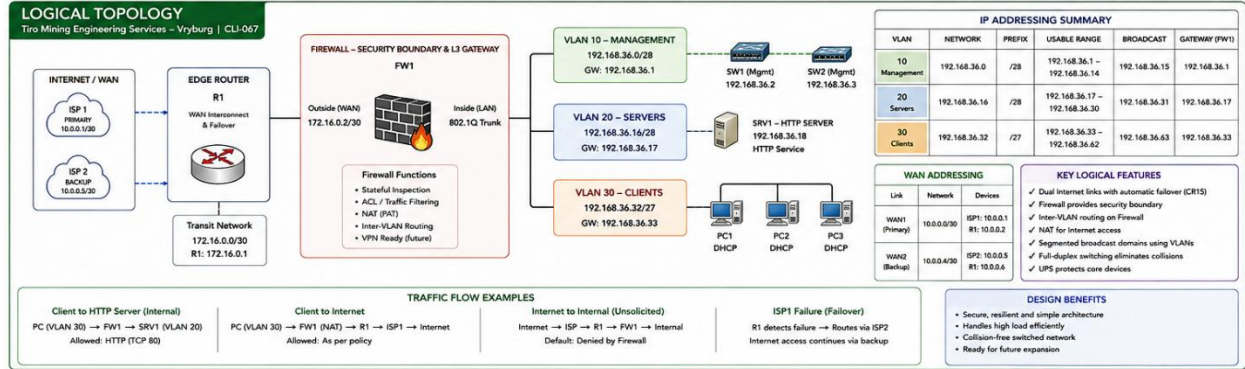
Source	Destination	Purpose
ISP1	R1 WAN1	Primary Internet
ISP2	R1 WAN2	Backup Internet
R1	FW1	WAN-to-LAN boundary
FW1	SW1	Internal LAN
SW1	SW2	VLAN trunk
SW1	SRV1	Server connection
SW2	PC1–PC3	Client connections

2.4 UPS Resilience

UPS supports R1, FW1, SW1 (and SW2 if capacity allows), ensuring uptime during load-shedding.

3. Logical Topology

3.1 VLAN Structure



VLAN	Purpose	Network	Gateway
10	Management	192.168.36.0/28	192.168.36.1
20	Servers	192.168.36.16/28	192.168.36.17
30	Clients	192.168.36.32/27	192.168.36.33

3.2 Routing & Traffic Control

- **Inter-VLAN Routing:** FW1 provides gateways for VLANs.
- **Firewall Policies:**
 - Permit internal clients → Internet.
 - Permit internal clients → HTTP server.
 - Deny unsolicited inbound traffic.
 - Permit authorised management traffic.

3.3 HTTP Server Placement

SRV1 resides in VLAN 20 with static IP **192.168.36.18**, accessible from VLAN 30 clients via inter-VLAN routing.

3.4 Dual-WAN Failover

- Normal: LAN → FW1 → R1 → ISP1 → Internet.
- Failover: LAN → FW1 → R1 → ISP2 → Internet. Tested by disabling ISP1 and verifying ISP2 routing.

3.5 Collision & Broadcast Management

- Switch ports = separate collision domains.
- Full-duplex Ethernet eliminates collisions.
- VLANs separate broadcast domains (Mgmt, Servers, Clients).

4. IP Addressing Plan

4.1 Assigned Address Block

The client has been allocated the private IPv4 block: **192.168.36.0/24**

This block is divided using **Variable Length Subnet Masking (VLSM)** to ensure efficient use of addresses, clear separation of functions, and scalability for future expansion.

4.2 VLSM Allocation:

VLAN	Purpose	Network	Prefix	Usage Range	Broadcast	Gateway
10	Management	192.168.36.0	/28	192.168.36.1–14	.15	192.168.36.1
20	Servers	192.168.36.16	/28	192.168.36.17–30	.31	192.168.36.17
30	Client's	192.168.36.32	/27	192.168.36.33–62	.63	192.168.36.33
-	Reserved	192.168.36.64	/26	192.168.36.65–126	.127	-

4.3 Device Address Assignments:

Device	VLAN	IP Address	Subnet Mask	Purpose
FW1 Mgmt	10	192.168.36.1	255.255.255.240	Management Gateway
SW1 Mgmt	10	192.168.36.2	255.255.255.240	Core Switch Mgmt
SW2 Mgmt	10	192.168.36.3	255.255.255.240	Access Switch Mgmt
FW1 Server GW	20	192.168.36.17	255.255.255.240	Server Gateway
SRV1 HTTP Server	20	192.168.36.18	255.255.255.240	Internal Web Service
FW1 Client GW	30	192.168.36.33	255.255.255.224	Client Gateway
PC1	30	DHCP	255.255.255.224	Client Workstation
PC2	30	DHCP	255.255.255.224	Client Workstation
PC3	30	DHCP	255.255.255.224	Client Workstation

4.4.1 WAN & Transit Networks:

Link	Network	Device Addressing
ISP1 Primary	10.0.0.0/30	ISP1: 10.0.0.1, R1 WAN1: 10.0.0.2
ISP2 Backup	10.0.0.4/30	ISP2: 10.0.0.5, R1 WAN2: 10.0.0.6
R1-FW1 Transit	172.16.0.0/30	R1: 172.16.0.1, FW1: 172.16.0.2

4.4.2 DHCP Configuration:

Parameter	Value
Network	192.168.36.32/27
Default Gateway	192.168.36.33
Pool Range	192.168.36.34–62
DNS Server	8.8.8.8 / Internal

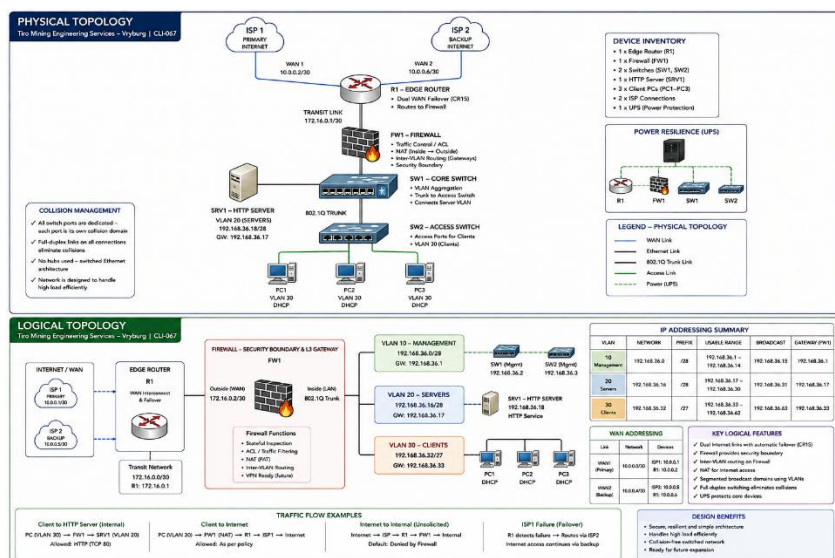
4.5 DHCP Plan

- **Network:** 192.168.36.32/27 (Client VLAN)
- **Gateway:** 192.168.36.33
- **Pool:** 192.168.36.34–62
- **Exclusions:** Gateway and infrastructure addresses reserved as static.

4.6 Addressing Design Rationale

- **Segmentation:** VLANs separate management, servers, and clients into distinct broadcast domains.
- **Predictability:** Static IPs for infrastructure and server resources ensure consistency.
- **Flexibility:** DHCP for clients simplifies management and supports dynamic scaling.
- **Resilience:** WAN and transit networks are isolated from the internal /24 block, ensuring clean separation of external and internal routing.
- **Scalability:** Reserved subnets allow for future expansion without redesign

4.7 Design Summary



The proposed architecture is: 2 ISPs → 1 Edge Router → 1 Firewall → 2 Switches → 3 PCs + 1 HTTP Server

It satisfies all requirements by providing:

- Internal HTTP/Web server hosting.
- Dual ISP resilience (CR15).
- Firewall-based traffic control.
- VLAN segmentation for management, servers, and clients.
- Collision and broadcast separation.
- UPS resilience for load-shedding.
- Efficient use of the 192.168.36.0/24 block with VLSM.
- Minimal device count suitable for Packet Tracer.

5 Initial Github Repository:

“ https://github.com/Samkelokhza/Computer_Networks-Tiro_Mining_Network_Project”